

16.1 In Examples 16.2 and 16.4, we presented the linear probability and probit model estimates using an example of transportation choice. The logit model for the same example is $P(AUTO = 1) = \Lambda(\gamma_1 + \gamma_2 DTIME)$, where $\Lambda(\bullet)$ is the logistic cdf in equation (16.7). The logit model parameter estimates and their standard errors are

$$\begin{array}{l} \hat{\gamma}_1 + \hat{\gamma}_2 DTIME = -0.2376 + 0.5311 DTIME \\ (se) \quad (0.7505) \quad (0.2064) \end{array}$$

- Calculate the estimated probability that a person will choose automobile transportation given that $DTIME = 1$.
- Using the probit model results in Example 16.4, calculate the estimated probability that a person will choose automobile transportation given that $DTIME = 1$. How does this result compare to the logit estimate? [Hint: Recall that Statistical Table 1 gives cumulative probabilities for the standard normal distribution.]
- Using the logit model results, compute the estimated marginal effect of an increase in travel time of 10 minutes for an individual whose travel time is currently 30 minutes longer by bus (public transportation). Using the linear probability model results, compute the same marginal effect estimate. How do they compare?
- Using the logit model results, compute the estimated marginal effect of a decrease in travel time of 10 minutes for an individual whose travel time is currently 50 minutes longer by driving. Using the probit results, compute the same marginal effect estimate. How do they compare?

$$a. P(AUTO=1) = \Lambda(\gamma_1 + \gamma_2 DTIME) = \frac{1}{1 + e^{-(\gamma_1 + \gamma_2 DTIME)}}$$

$$Z = -0.2376 + 0.5311(1) = 0.2935$$

$$P = \frac{1}{1 + e^{-0.2935}} = \frac{1}{1.9456} \approx 0.5129$$

$$b. P(AUTO=1) = \Phi(-0.0644 + 0.3 DTIME)$$

$$Z = -0.0644 + 0.3(1) = 0.2356$$

$$P = \Phi(0.2356) \approx 0.5931$$

機率相近(51.3% vs 59.3%) 在機率接近0.5的中央區域

Probit & Logit 模型算出的預測結果幾乎沒差異。

$$d. Z = -0.2376 + 0.5311(-5) = -2.8931$$

$$P = \Lambda(-2.8931) = \frac{1}{1 + e^{2.8931}} \approx 0.0525$$

$$ME = 0.0525(1 - 0.0525) \times 0.5311 \approx 0.0264$$

e Probit

$$Z = -0.0644 + 0.3(-5) = -1.5644$$

$$P = \Phi(-1.5644) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(-1.5644)^2} \approx 0.1173$$

$$ME = \phi(-1.5644) \times \beta_2 = 0.1173(0.3) \approx 0.0352$$

$$ME_{logit} \approx 2.44\%$$

$$ME_{probit} \approx 3.52\%$$

$DTIME = -5$ 分佈在極端的尾部
不論是 Probit or Logit 計算出的 ME 都很小

$$c. LPM \quad P = \alpha_1 + \alpha_2 DTIME$$

$$\frac{\partial P}{\partial DTIME} = \alpha_2 = 0.0703$$

Logit 的 ME 是動態的，取決於 $DTIME$

而 LPM 的 ME 不管 $DTIME$ 變動都為固定常數。

16.4 In Example 16.3, we illustrate the calculation of the likelihood function for the probit model in a small example. In this exercise, we will repeat that example using logit instead of probit. The logit model for the same example is $P(y = 1) = \Lambda(\gamma_1 + \gamma_2 x)$, where $\Lambda(\bullet)$ is the logistic cdf in equation (16.7). The maximum likelihood estimates of the parameters are $\hat{\gamma}_1 + \hat{\gamma}_2 x = -1.836 + 3.021x$. The maximized value of the log-likelihood function is -1.612 .

- Calculate the probability that $y = 1$ if $x = 1.5$, given the values of the maximum likelihood estimates.
- Using the threshold 0.5 and the result in part (a), predict the value of y if $x = 1.5$, the first observation, given the values of the maximum likelihood estimates. Compare your prediction to the actual outcome $y = 1$ in the first observation.
- Calculate the value of the likelihood function, illustrated in equation (16.14) but substituting equation (16.17) in place of $\Phi(\bullet)$ and using the given $N = 3$ data pairs, if the parameter values are $\gamma_1 = -1$ and $\gamma_2 = 2$. Compare this value to the value of the likelihood function evaluated at the maximum likelihood estimates. Which is larger?
- For the logit model, the value of the likelihood function (16.14), with $\Lambda(\bullet)$ in place of $\Phi(\bullet)$, will always be between zero and one. True or false? Explain.
- For the logit model, the value of the log-likelihood function (16.15), with $\Lambda(\bullet)$ in place of $\Phi(\bullet)$, will always be negative. True or false? Explain.

$$a. \quad z = -1.836 + 3.021(1.5) = 2.6955$$

$$p(y=1|x=1.5) = \frac{1}{1 + e^{-2.6955}} \approx 0.9368$$

$$b. \quad \begin{cases} p > 0.5 \rightarrow y=1 \\ p \leq 0.5 \rightarrow y=0 \end{cases} \quad \because p = 0.9368 > 0.5 \quad \therefore \hat{y} = 1$$

$$c. \quad z_i = -1 + 2x_i$$

$$\textcircled{1} \quad y_1 = 1, x_1 = 1.5$$

$$z_1 = -1 + 2(1.5) = 2$$

$$p_1 = \Lambda(2) = \frac{1}{1 + e^{-2}} \approx 0.8808$$

$$\textcircled{2} \quad y_2 = 1, x_2 = 0.6$$

$$z_2 = -1 + 2(0.6) = 0.2$$

$$p_2 = \Lambda(0.2) = \frac{1}{1 + e^{-0.2}} \approx 0.5498$$

$$\textcircled{3} \quad y_3 = 0, x_3 = 0.9$$

$$z_3 = -1 + 2(0.9) = 0.4$$

$$p_3 = 1 - \Lambda(0.4) = 1 - \frac{1}{1 + e^{-0.4}} \approx 0.4013$$

$$L(-1, 2) = p_1 \times p_2 \times p_3 = (0.8808)(0.5498)(0.4013) \approx 0.1943$$

$$\ln(L) = -1.612$$

$$L_{MLE} = e^{-1.612} \approx 0.1995$$

MLE 的 Likelihood function 較大
最大似估計法求出的那組參數
一定會讓似函數最大。

d. True

Likelihood function 是各樣本發生機率的乘積，logit 算出來的機率都是 0~1 多個 (0,1) 相乘結果必落在 0~1 區間內。

e. True

$\ln(L)$ 由 d 小題已知 $0 < L < 1$

任何 0~1 的正數，取對數值必為負值

$$\ln(1) = 0, \ln 0.5 = -0.693 \dots$$